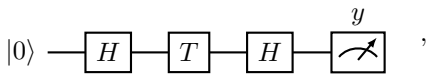
Toy examples for verifying and explaining PACTS

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1 Regular normal form procedure

1.1 Toy Example 1

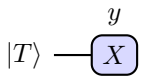
Our first example is the following simple one-qubit circuit with a single T gate,



(1)

which has an output probability distribution $p(y=0) = 1/2 + \sqrt{2}/4 \approx 0.8536$ and $p(y=1) = 1/2 - \sqrt{2}/4 \approx 0.1464$.

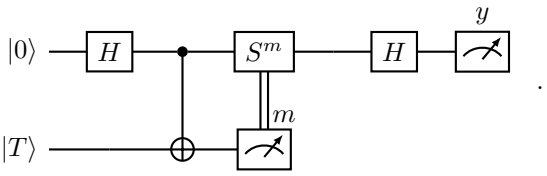
A simple rewriting of the circuit shows that an equivalent computation is:



(2)

where the blue box with rounded edges is used to denote a quantum measurement of X. This notation is used throughout. The computation depicted in Eq. (2) corresponds to the Pauli-based computation (PBC) that outputs samples from the output distribution of the quantum circuit in Eq. (1).

In this case, finding the corresponding PBC is particularly trivial and we need not carry out our general normal form procedure. However, as we believe doing so might be instructive and pedagogical, in what follows, we work out the example explicitly in that way. We start by rewriting the quantum circuit in Eq. (1) using magic-state injection:

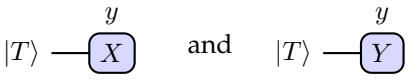


Once this is done, we proceed as described in our paper. First, we backpropagate the measurement associated with the T gadget until it arrives at the beginning of the circuit as an $X \otimes Z$ Pauli measurement. This is clearly identified as an anticommuting measurement that can, therefore, be classically processed. Because it corresponds to a gadget measurement, we can postselect its outcome to $m = 0$ without loss of generality. We then store information about the Clifford unitary

$$V = \frac{Z \otimes I + X \otimes Z}{\sqrt{2}},$$

which needs to be used to update upcoming Pauli measurements. Finally, we backpropagate the readout measurement, which arrives at the beginning of the quantum circuit as $Z \otimes X$. Updating this Pauli measurement under V leaves it unchanged, so this is the next Pauli to be considered. Since $Z \otimes X$ clearly commutes with $Z \otimes I$, it constitutes the first measurement that must be performed in quantum hardware. Importantly, we can reduce the measurement to what happens in the magic register. Thus, in the end, we retrieve exactly the computation depicted in Eq. (2).

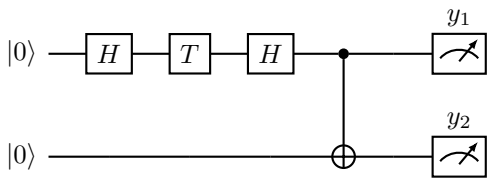
Here, we point out that, in previous descriptions and implementations of the PBC procedure (e.g. [1–3]), postselection of classically decidable gadget measurements was not performed. This was one of the factors hindering the earlier establishment of a PBC normal form. In the particular case we are now considering, previous solutions established that two distinct PBCs would have to be carried out [3]:



Here, we see that this is unnecessary: Performing the computation in Eq. (2) suffices to produce samples from the correct output distribution. **PACTS** precisely captures this observation.

1.2 Toy Example 2

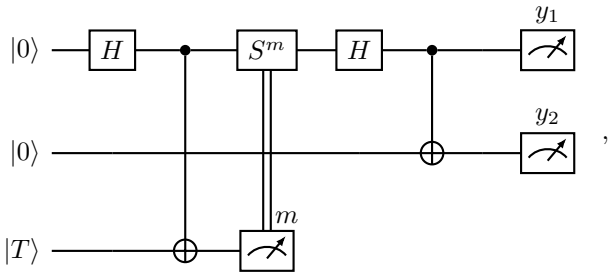
Our second example is the following quantum circuit:



, (3)

which has an output probability distribution given by $p(00) = 1/2 + \sqrt{2}/4 \approx 0.8536$ and $p(11) = 1/2 - \sqrt{2}/4 \approx 0.1464$.

Since this quantum circuit has only one T gate, we know that it will be simulable by a PBC on a single qubit. We will use our proposed normal form procedure to find that PBC. As usual, the starting step is to transform the T gate into a T gadget, so that we get the following adaptive quantum circuit:



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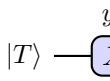
Backpropagating the gadget measurement leads to $X \otimes I \otimes Z$, which clearly anticommutes with $Z \otimes I \otimes I$. Thus, we classically fix $m = 0$ and the unitary

$$V = \frac{Z \otimes I \otimes I + X \otimes I \otimes Z}{\sqrt{2}} .$$

We can now backpropagate the readout measurements:

$$Z \otimes I \otimes I \mapsto Z \otimes I \otimes X$$
$$I \otimes Z \otimes I \mapsto Z \otimes Z \otimes X .$$

Both measurements commute with the stabilizers of $|0\rangle^{\otimes 2}$. Thus, we can take the first readout measurement and perform it in a quantum computer. The second readout measurement is clearly dependent on the first. In the end, this means that the quantum circuit in Eq. (3) can be simulated by the following PBC:

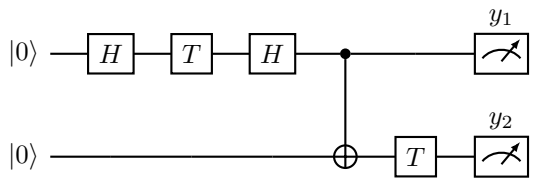


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together with classical side-processing that establishes $y_2 = y_1$. Again, unlike prior works, which did not establish a unique PBC solution, **PACTS** returns the solution above and outputs samples from the correct distribution.

1.3 Toy Example 3

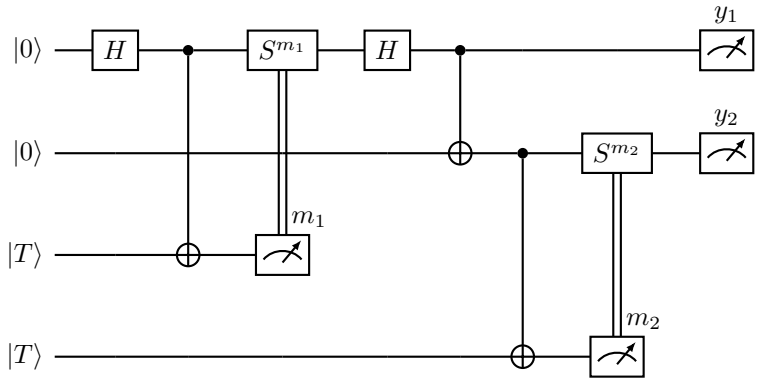
Our next example is the following quantum circuit:



(4)

It is clear that the T gate acting on the second qubit has no impact on the outcome of the computation, so that this circuit is perfectly equivalent (from a computational standpoint) to the one in Eq. (3). Here, we add this second T gate with the sole purpose of (artificially) manufacturing an example that leads to a two-qubit PBC (rather than a simple one-qubit one).

We construct the corresponding adaptive Clifford circuit by replacing each T gate by a magic state injection gadget:



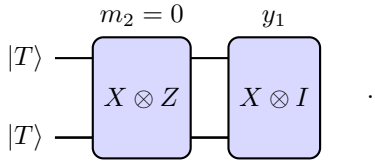
As usual, we start by backpropagating the measurement associated with the first gadget. This yields $I\otimes I\otimes Z\otimes I \mapsto X\otimes I\otimes Z\otimes I$, which corresponds to an anticommuting Pauli. Thus, we fix $m_1 = 0$ and store information concerning the Clifford unitary:

$$V = \frac{Z\otimes I\otimes I\otimes I + X\otimes I\otimes Z\otimes I}{\sqrt{2}}\,.$$

We move on with the second gadget measurement, which after backpropagation through the quantum circuit yields: $I\otimes I\otimes I\otimes Z \mapsto Z\otimes Z\otimes X\otimes Z$. This Pauli remains unchanged under update by V thus, in the end, it gets recognized as the first quantum measurement that needs to be performed in the quantum hardware. Following our normal form procedure, to create the zero branch we assume $m_2 = 0$. The circuit is now fixed and we can backpropagate both readout measurements through the circuit obtaining:

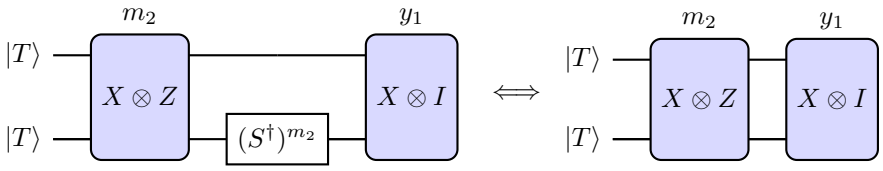
$$\begin{aligned} Z\otimes I\otimes I\otimes I &\mapsto Z\otimes I\otimes X\otimes I \\ I\otimes Z\otimes I\otimes I &\mapsto Z\otimes Z\otimes X\otimes I\,. \end{aligned}$$

The first readout is identified as the second quantum measurement that must be performed in quantum hardware while the second readout is dependent on the first ensuring that $y_2 = y_1$. In the end, this fixes the first branch of the PBC as:



Our normal form theorem informs us that the second branch of the PBC (obtained when m_1) is exactly the same as

the zero branch:



(5)

For completeness, let us show that the outcomes y_1 and y_2 follow the correct output distribution. The state of the system after the first quantum measurement is:

$$\left|\psi_1^{(\pm)}\right\rangle \propto \frac{I\otimes I \pm X\otimes Z}{2}|T\rangle^{\otimes 2} = \frac{1}{4}\left[(1\pm e^{i\pi/4})|00\rangle + (e^{i\pi/4}\mp i)|01\rangle + (e^{i\pi/4}\pm 1)|10\rangle + (i\mp e^{i\pi/4})|11\rangle\right] .$$

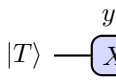
From the unnormalized results it is easy to check that $p(m_2=0)=p(m_2=1)=0.5$ as need be.

Measuring $X\otimes I$ on the (normalized) quantum state $\left|\psi_1^{(+)}\right\rangle$ yields:

$$\begin{aligned} y_1=0: \quad |\psi_2\rangle &\propto \frac{I\otimes I + X\otimes I}{2}\left|\psi_1^{(+)}\right\rangle = \frac{1+\sqrt{2}+i}{4}\left(|00\rangle+|10\rangle\right) \\ y_2=0: \quad |\psi_2'\rangle &\propto \frac{I\otimes I - X\otimes I}{2}\left|\psi_1^{(+)}\right\rangle = \frac{1+i(1-\sqrt{2})}{4}\left(|01-|11\rangle\right) . \end{aligned}$$

From the unnormalized results we obtain that $p(y_1=0)=1/2+\sqrt{2}/4$ and $p(y_1=1)=1/2-\sqrt{2}/4$. Combined with the classical information that $y_2=y_1$, this guarantees the correct output distribution. Similar calculations ensure that the same output distribution holds for the second branch ($m_2=1$).

Free virtual qubits.— This is an example where **PACTS** will successfully detect that one of the magic qubits can be freely removed. From Eq. (5), we see that the second qubit is only ever acted on by I or Z. Thus, it can be removed from the computation. This is done by noting that, in such a case, having $|T\rangle$ or having $|+\rangle$ is perfectly equivalent. Thus, $X\otimes Z$ becomes an anti-commuting measurement. We can post-select $m_2=0$, which will thus lead to the following PBC:

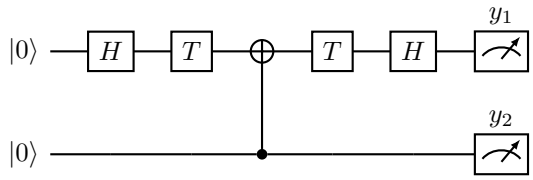


That is, we obtain exactly the same result as for Toy Example 2. This makes sense, since, as commented before, the T gate was artificially added. The normal form now allows us to identify such 'useless' T gates. Something that was not possible before.

1.4 Toy Example 4

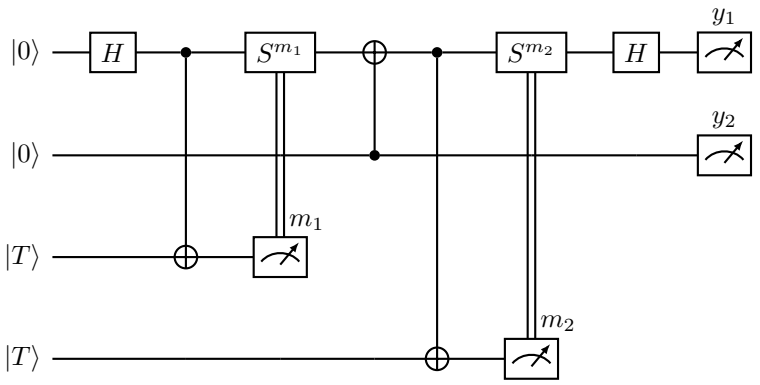
Our first three examples consist of circuits that are X-invariant, i.e., circuits whose output distribution remains unchanged (up to bit re-interpretation) when the input state is altered into a different computational basis state. Concretely, if in the first example we took the input of the quantum circuit to be $|1\rangle$ rather than $|0\rangle$, then we would have: $p(0) = 1/2 - \sqrt{2}/4 \approx 0.1464$ and $p(1) = 1/2 + \sqrt{2}/4 \approx 0.8536$. Similar observations apply to the second and third examples.

We now consider a quantum circuit which is not X-invariant:



. (6)

The output distribution of this quantum circuit is $p(00) = p(10) = 0.5$. Let us see what the corresponding PBC looks like under our normal-form theorem. As usual, we transform the T gates into T gadgets:



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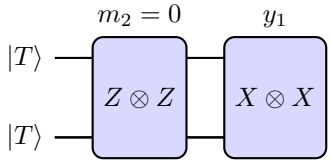
Backpropagating the first gadget measurement, we obtain: $X \otimes I \otimes Z \otimes I$, which is an anticommuting measurement. We classically postselect $m_1 = 0$ and store information about the Clifford unitary:

$$V = \frac{Z \otimes I \otimes I \otimes I + X \otimes I \otimes Z \otimes I}{\sqrt{2}} .$$

Next, we backpropagate the second gadget measurement, obtaining $Z \otimes Z \otimes Z \otimes Z$ (already after update under conjugation by V). As this is a quantum measurement that clearly commutes with all the stabilizers of $|0\rangle^{\otimes 2}$ and is independent of them, we establish $Z \otimes Z$ as the first quantum measurement. Let us fix $m_2 = 0$. This allows us to backpropagate the two readout measurements:

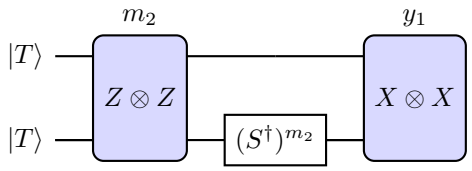
$$\begin{aligned} Z \otimes I \otimes I \otimes I &\mapsto Z \otimes I \otimes X \otimes X \\ I \otimes Z \otimes I \otimes I &\mapsto I \otimes Z \otimes I \otimes I \quad . \end{aligned}$$

Hence, the first readout is identified as a measurement that needs to be performed in quantum hardware; specifically, we perform the measurement $X \otimes X$ in the magic register. On the other hand, the second readout measurement is a dependent measurement that can be classically determined: It deterministically yields $y_2 = 0$. This establishes the following as the zero branch of this PBC:



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Generally, our normal form theorem establishes the following as the general structure of this PBC:



. (7)

Thus, this PBC has two branches. It is not hard to show that, in each of these branches, $p(y_1 = 0) = p(y_1 = 1) = 0.5$. For completeness, we explicitly prove this claim. Assuming the first quantum measurement yields $m_2 = 0$, then, the state of the quantum system after such measurement is:

$$|\psi_1\rangle \propto \frac{I \otimes I + Z \otimes Z}{2} |T\rangle^{\otimes 2} = \frac{1}{2}(|00\rangle + i\,|11\rangle)\,.$$

Then, performing the measurement $(X \otimes X)$ on $|\psi_1\rangle$ leads to:

$$|\psi_2\rangle \propto \frac{I \otimes I \pm X \otimes X}{2} |\psi_1\rangle = \frac{I \otimes I \pm X \otimes X}{2\sqrt{2}}(|00\rangle + i\,|11\rangle) = \frac{(1 \pm i)}{2} \frac{|00\rangle \pm |11\rangle}{\sqrt{2}} \implies p(y_1 = 0) = p(y_1 = 1) = 0.5\,.$$

Assuming the first quantum measurement yields $m_2 = 1$, then, the state of the quantum system after such measurement is:

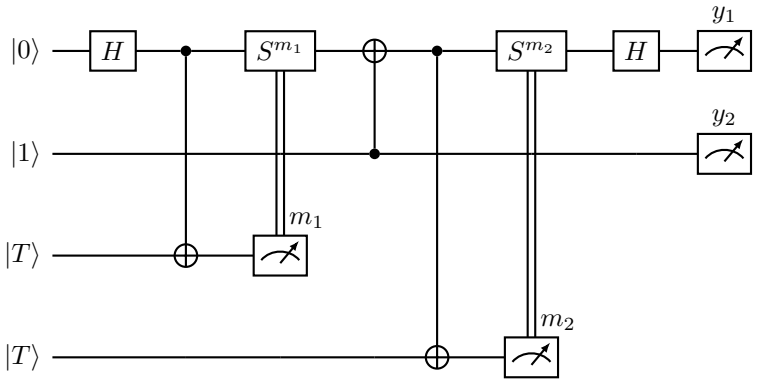
$$|\psi_1'\rangle \propto \frac{I \otimes I - Z \otimes Z}{2} |T\rangle^{\otimes 2} = \frac{e^{i\pi/4}}{2}(|01\rangle + |10\rangle)\,.$$

Then, performing the measurement $(X \otimes Y)$ on $|\psi_1\rangle$ leads to:

$$|\psi_2'\rangle \propto \frac{I \otimes I \pm X \otimes Y}{2} |\psi_1'\rangle = \frac{(-i)^{y_1}}{2} [i\,|01\rangle + (-1)^{y_1}\,|10\rangle] \implies p(y_1 = 0) = p(y_1 = 1) = 0.5\,.$$

PACTS successfully constructs the PBC tree in Eq. (2) and outputs the correct distribution $p(00) = p(10) = 0.5$.

Since this circuit is not X-invariant, we take the opportunity to study what happens if the second qubit in Eq. (4) is initialized in state $|1\rangle$, rather than $|0\rangle$. In this case, one can show that the output distribution of the quantum circuit is $p(01) = 1$. What is the corresponding PBC in this case? Our starting point now is the quantum circuit:



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Following the usual procedure, we find that, just as before, the first gadget measurement is $X \otimes I \otimes Z \otimes I$, which is an anticommuting measurement. We classically postselect $m_1 = 0$ and store information about the Clifford unitary:

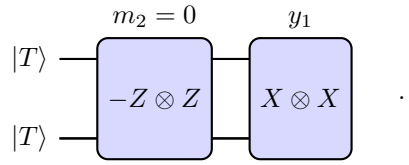
$$V = \frac{Z \otimes I \otimes I \otimes I + X \otimes I \otimes Z \otimes I}{\sqrt{2}}\,.$$

Next, we backpropagate the second gadget measurement, obtaining $Z \otimes Z \otimes Z \otimes Z$ (already after update under conjugation by V). As this is a quantum measurement that clearly commutes with all the stabilizers of $|0\rangle^{\otimes 2}$ and is independent of them, we establish $-Z \otimes Z$ as the first quantum measurement. Again, fixing $m_2 = 0$, allows us to

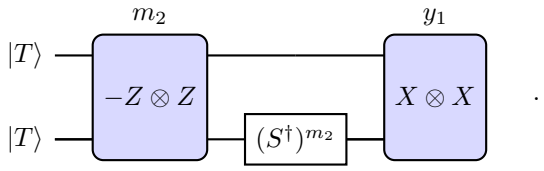
backpropagate the two readout measurements:

$$Z \otimes I \otimes I \otimes I \mapsto Z \otimes I \otimes X \otimes X$$
$$I \otimes Z \otimes I \otimes I \mapsto I \otimes Z \otimes I \otimes I \quad .$$

The first readout is identified as a measurement that needs to be performed in quantum hardware; specifically, we perform the measurement $X \otimes X$ in the magic register. On the other hand, the second readout measurement is a dependent measurement that can be classically determined: It deterministically yields $y_2 = 1$. This establishes the following as the zero branch of this PBC:



Following our normal formal theorem, the general structure of this PBC is:



While seemingly similar to the PBC in Eq. (7), this computation is fundamentally different; in fact, straightforward calculations show that each branch has a probability distribution: $p(0) = 1$. For completeness, we show this explicitly. Suppose $m_2 = 0$; then, the state of the system after the first quantum measurement is:

$$|\psi_1\rangle \propto \frac{I \otimes I - Z \otimes Z}{2} |T\rangle^{\otimes 2} = \frac{e^{i\pi/4}}{2}(|01\rangle + |10\rangle) .$$

Then, measuring $X \otimes X$ on this state yields:

$$y_1 = 0 : \quad |\psi_2\rangle \propto \frac{I \otimes I + X \otimes X}{2} |\psi_1\rangle = \frac{e^{i\pi/4}}{\sqrt{2}}(|01\rangle + |10\rangle) \Longrightarrow p(y_1 = 0) = 1 ,$$
$$y_1 = 1 : \quad |\psi_2\rangle \propto \frac{I \otimes I - X \otimes X}{2} |\psi_1\rangle = 0 \Longrightarrow p(y_1 = 1) = 0 .$$

Suppose $m_2 = 1$; then, the state of the system after the first quantum measurement is:

$$|\psi_1'\rangle \propto \frac{I \otimes I + Z \otimes Z}{2} |T\rangle^{\otimes 2} = \frac{1}{2}(|00\rangle + i\,|11\rangle) .$$

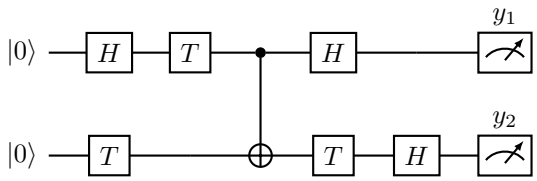
Then, measuring $X \otimes Y$ on this state yields:

$$y_1 = 0 : \quad |\psi_2'\rangle \propto \frac{I \otimes I + X \otimes Y}{2} |\psi_1'\rangle = \frac{1}{\sqrt{2}}(|00\rangle + i\,|11\rangle) \Longrightarrow p(y_1 = 0) = 1 ,$$
$$y_1 = 1 : \quad |\psi_2'\rangle \propto \frac{I \otimes I - X \otimes Y}{2} |\psi_1'\rangle = 0 \Longrightarrow p(y_1 = 1) = 0 .$$

We use **PACTS** to obtain samples from the output distribution of the quantum circuit in Eq. (6). We also change the input to the quantum circuit from $|00\rangle$ to $|01\rangle$ and verify that the outputs suitably follow the new distribution. For completeness, we test **PACTS** with inputs $|10\rangle$ and $|11\rangle$ for which the correct output distributions are $p(00) = p(10) = 0.5$ and $p(11) = 1$, respectively.

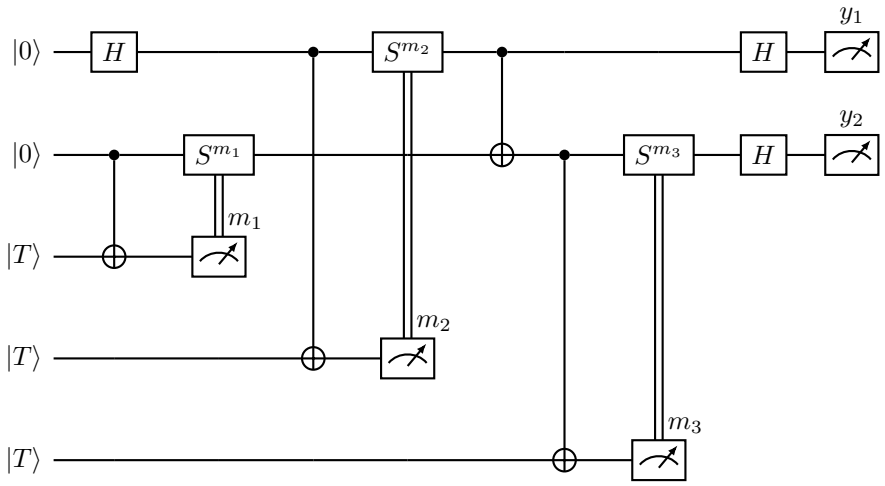
1.5 Toy Example 5

Here, we consider another quantum circuit that is not X-invariant:



. (8)

This quantum circuit has a uniform output distribution: $p(00) = p(01) = p(10) = p(11) = 0.25$. Let us compute the corresponding PBC and check that it precisely yields this output distribution. As usual, we start by constructing the adaptive Clifford circuit:



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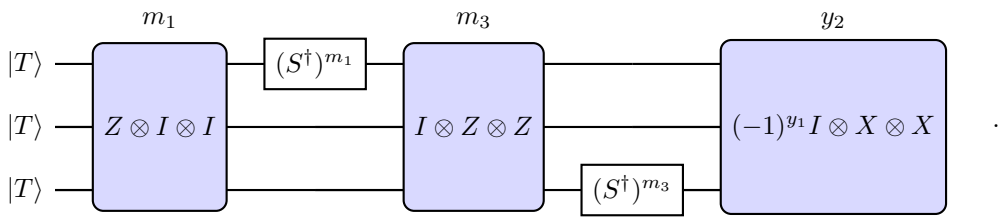
We start by backpropagating the first gadget measurement, for which we obtain: $I \otimes I \otimes Z \otimes I \otimes I \mapsto I \otimes Z \otimes Z \otimes I \otimes I$ which can easily be identified as a measurement to be performed in the quantum computer. Assuming $m_1 = 0$, we can then backpropagate the measurement associated with the second gadget: $I \otimes I \otimes I \otimes Z \otimes I \mapsto X \otimes I \otimes I \otimes Z \otimes I$, which is easily recognized as an anticommuting measurement whose outcome is decided classically. We postselect $m_2 = 0$ and store information about the Clifford unitary:

$$V_1 = \frac{Z \otimes I \otimes I \otimes I \otimes I + X \otimes I \otimes I \otimes Z \otimes I}{\sqrt{2}} .$$

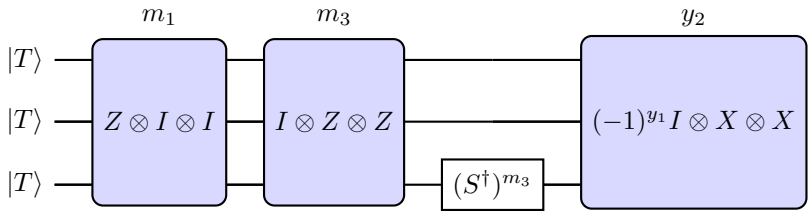
We follow this by the backpropagation of the third gadget measurement, which yields: $I \otimes I \otimes I \otimes I \otimes Z \mapsto Z \otimes Z \otimes I \otimes Z \otimes Z$ (already including the update under conjugation by V_1). This is clearly a measurement that needs to be done in the quantum hardware. Assuming $m_3 = 0$, we can now backpropagate the two readout measurements to complete our understanding of the zero branch. Let us start with the first: $Z \otimes I \otimes I \otimes I \otimes I \mapsto Z \otimes X \otimes X \otimes X \otimes I$, which clearly anticommutes with $I \otimes Z \otimes I \otimes I \otimes I$. Thus, it is clear that y_1 is decided via cointoss so that $y_1 \in_R \{0, 1\}$, fixing a second Clifford unitary:

$$V_2(y_1) = \frac{I \otimes Z \otimes I \otimes I \otimes I + (-1)^{y_1} Z \otimes X \otimes X \otimes X \otimes I}{\sqrt{2}} .$$

Let us finally backpropagate the second readout to the beginning of the quantum circuit: $I \otimes Z \otimes I \otimes I \otimes I \mapsto (-1)^{y_1} Z \otimes Z \otimes I \otimes X \otimes X$. Hence, this means that the PBC computation looks as follows:



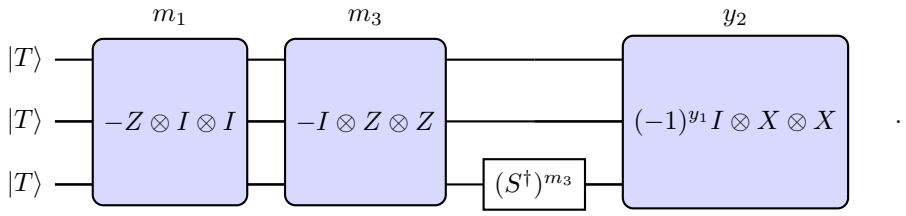
Since the conjugation by $(S^\dagger)^{m_1}$ has no effect on upcoming measurements, this PBC can be simplified to:



(9)

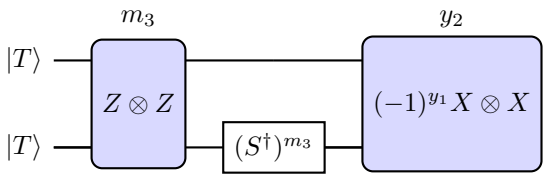
which indicates a PBC with two branches. We note that the same computation can be done with only two qubits, since the first magic qubit is not entangled with the other two.

Again, since this is not an X-invariant quantum circuit, if we change the input computational basis state, then the output probability changes form. For the input $|01\rangle$ the output probability distribution is $p(00) = p(11) = 0.5$. For this particular case, the PBC looks very similar. Specifically, we get:



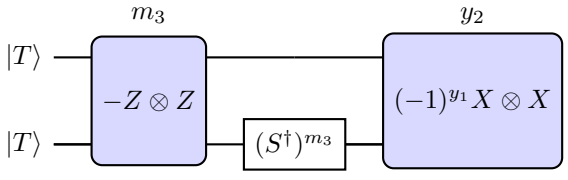
Free virtual qubits.— This is another example where **PACTS** will also successfully detect that one of the magic qubits can be freely removed. From Eq. (9), we see that the first qubit is only ever acted on by I or Z. Thus, it can be removed from the computation, leading to the following PBCs:

Input: $|00\rangle$



, and

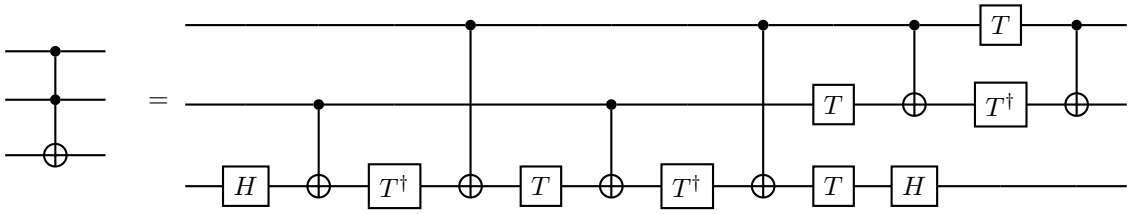
Input: $|01\rangle$



.

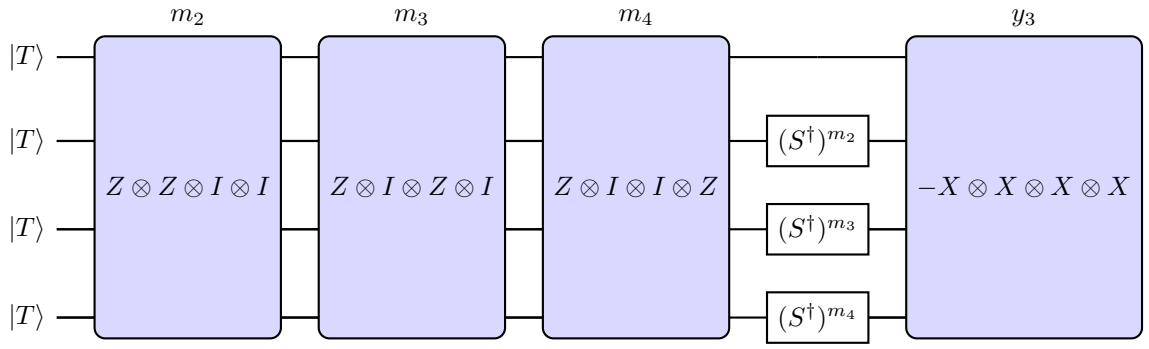
1.6 Toy Example 6 — Toffoli gate

Here, we test **PACTS** with the Clifford+T circuit implementing the Toffoli gate:



Since this example involves 7 T gates, we forego the explicit calculation of the PBC tree and instead focus on the output distribution returned by **PACTS** for each of the 8 different computational basis states. We see that **PACTS** returns the correct output in each case.

For the input $|000\rangle$, the PBC obtained is the following:



. (10)

From the files output by **PACTS**, we further understand that the outcomes y_1 and y_2 are classical and deterministic: $y_1 = y_2 = 0$.

1.7 Interesting random quantum circuits

In addition to the toy examples in the previous sections, we also manufactured four different random quantum circuits with non-trivial structures and distributions. This serves to verify that **PACTS** is working even for more complex, intricate structures. The exact output probability distributions were calculated using Qiskit. Here, we write down these distributions and verify the output of **PACTS** against them.

Rand_2q_20g_0.— This circuit has a probability distribution given by $p(00) = p(11) \approx 0.4268$ and $p(01) = p(10) \approx 0.0732$.

Rand_2q_20g_1.— The output distribution of this quantum circuit is $p(00) \approx 0.1250$, $p(01) \approx 0.0214$, and $p(10) = p(11) \approx 0.4268$.

Rand_3q_30g_0.— The output distribution of this quantum circuit is $p(000) = p(001) = p(100) = p(101) \approx 0.2134$ and $p(010) = p(011) = p(110) = p(111) \approx 0.0366$.

Rand_4q_20g_0.— This circuit has an output probability distribution given by: $p(0000) = p(0010) = p(0101) = p(0111) = p(1001) = p(1011) = p(1100) = p(1110) \approx 0.1067$ and $p(0001) = p(0011) = p(0100) = p(0110) = p(1000) = p(1010) = p(1101) = p(1111) \approx 0.0183$.

All results obtained with **PACTS** are compatible with the output distributions presented above.

1.8 Grover's algorithm

In our repository, we also provide a small circuit implementing Grover's algorithm. **PACTS** can be run on this circuit to provide the PBC tree. For pedagogical purposes, we leave the explicit calculation of this example as an exercise to the interested user.

2 Simulating virtual qubits

If we want to simulate a certain number k of virtual qubits, **PACTS** does the following: For a t-qubit PBC, decompose $|T\rangle\langle T|^{\otimes k}$ into a linear combination of stabilizer states. For each term in the decomposition, find the corresponding PBC normal form. **PACTS** uses the following decomposition:

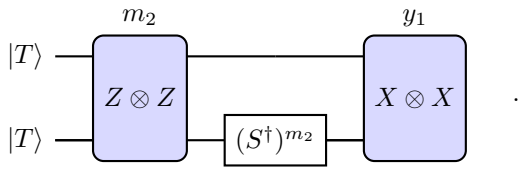
$$|T\rangle\langle T| = \frac{1}{2}|+\rangle\langle +| + \frac{1-\sqrt{2}}{2}|-\rangle\langle -| + \frac{1}{\sqrt{2}}|+_i\rangle\langle +_i| = \frac{1}{2}H|0\rangle\langle 0|H + \frac{1-\sqrt{2}}{2}ZH|0\rangle\langle 0|HZ + \frac{1}{\sqrt{2}}SH|0\rangle\langle 0|HS^\dagger ,$$

and, for simplicity, when considering k qubits, it takes the decomposition of $|T\rangle\langle T|^{\otimes k}$ to be the tensor product of the decomposition above.

Here, we illustrate the idea with a couple of simple toy examples.

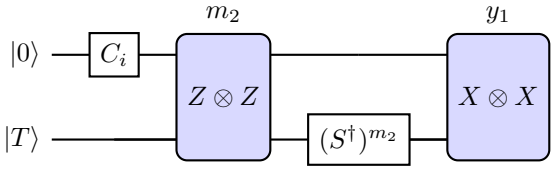
2.1 Toy Example 4

In Sec. 1.4, we saw that, for the circuit input state $|00\rangle$, the PBC normal form for this toy example is given by:



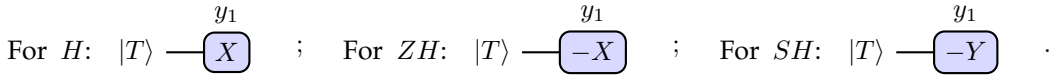
.

Suppose we now want to remove one qubit (equivalently, simulate one virtual qubit). Then, we can do the following:



,

where $\{C_i\}_{i=1}^3 \equiv \{H, ZH, SH\}$. Thus, for each of these three values, we can calculate a new PBC normal form. Specifically, it is not hard to show the following results:



.

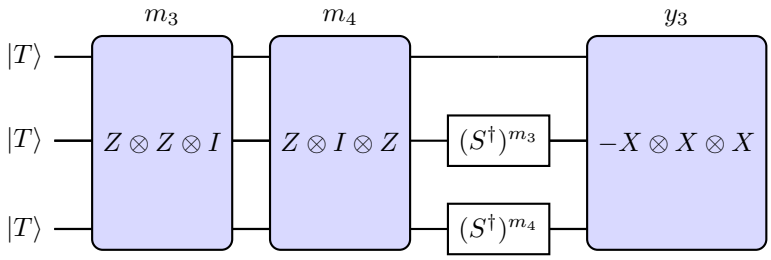
Thus, provided that we sample each of the three possibilities above with the correct probability and that the suitable number of samples is chosen, we will be able to estimate $p(y_1)$ with the desired precision [1].

Note that if, instead of estimating $p(y_1)$, we were interested in estimating $p(y_2)$, the result would be very different. In fact, it is easy to see that in that case no quantum computation is needed. This probability can be efficiently estimated by purely classical means. **PACTS** flags that rather easily.

2.2 Toy Example 6 — Toffoli gate

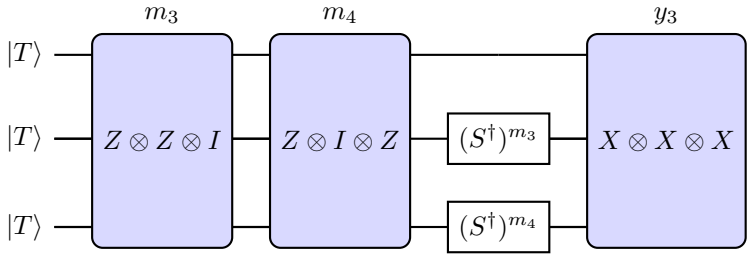
Suppose we would like to estimate $p(y_3)$ with a certain precision, while simulating 1 virtual qubit. Then, starting from Eq. (10), we see that there are three different PBC normal forms, which need to be sampled appropriately:

For H :



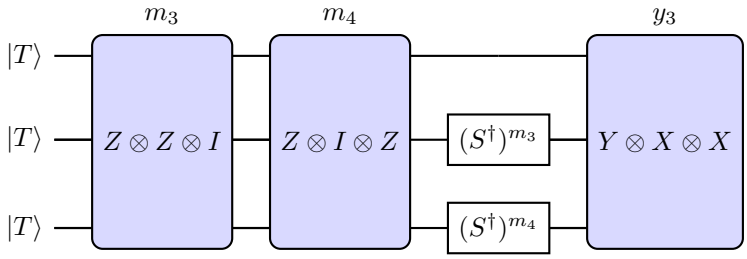
;

For ZH :



;

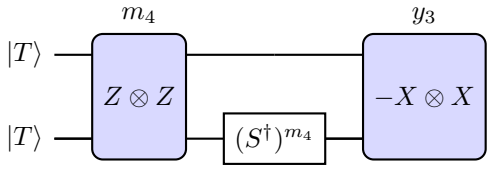
For SH :



.

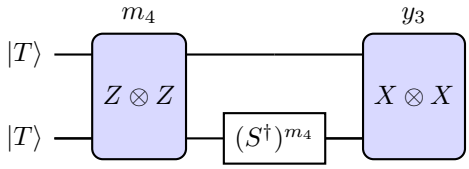
We can also simulate 2 virtual qubits, in which case we will have to sample one out of four different PBCs, following the appropriate probability distribution:

For $H \otimes H$ and $ZH \otimes ZH$:



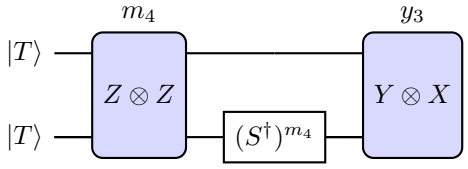
;

For $H \otimes ZH$, $ZH \otimes H$ and , $SH \otimes SH$:



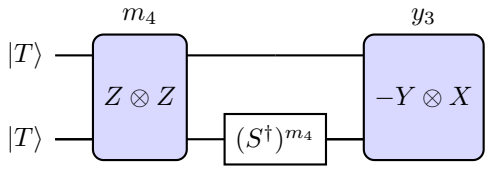
;

For $H \otimes SH$ and $SH \otimes H$:



;

For $ZH \otimes SH$ and $SH \otimes ZH$:



.

We could also do the same for 3 virtual qubits, but the idea would be the same, so we abstain from doing so.

We note that, for this example, estimating $p(y_1)$ and $p(y_2)$ requires no quantum resources, that is, it can be done purely classically, as both y_1 and y_2 are deterministically equal to zero. This is not hard to see from Eq. (10). This is reflected by the outcome of **PACTS**, signaling a fully classical computation.

References

[1] F. C. R. Peres and E. F. Galvão. "Quantum circuit compilation and hybrid computation using Pauli-based computation". In: *Quantum* 7 (Oct. 2023), p. 1126. ISSN: 2521-327X.

[2] F. C. R. Peres and E. F. Galvão. "Reducing depth and measurement weights in Pauli-based computation". In: *Phys. Rev. A* 112 (6 Dec. 2025), p. 062604.

[3] F. C. R. Peres. *CompHybPBC*. https://github.com/fcrperes/CompHybPBC. 2023.